

MANAGERIAL ACCOUNTING (135)

—Post-secondary—

STATE – 2019

Multiple Choice & Short-Answer Section:

Multiple Choice (15 @ 2 points each) _____ (30 points)

Short-Answers (12 @ 4 points each) _____ (48 points)

Production Portion:

Job 1: Budgeting Scenarios _____ (100 points)

Job 2: Variance Analysis _____ (96 points)

TOTAL POINTS _____ (***274 points***)

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

Property of Business Professionals of America.
May be reproduced only for use in the Business Professionals of America
Workplace Skills Assessment Program competition.

General Instructions

You have been hired as a Financial Assistant and will be keeping the accounting records for Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting and other financial services for clients. You will complete jobs for Professional Business Associates' own accounting records, as well as for clients.

Unless otherwise instructed, round all answers to 2 decimal places.

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question.

- 1) Which type of cost is fixed on a per unit basis but varies in direct proportion to changes in activity?
 - a) Fixed Cost
 - b) Variable Cost
 - c) Curvilinear Cost
 - d) Step-wise Cost
- 2) Which type of cost does *not* change as activity changes?
 - a) Mixed Cost
 - b) Variable Cost
 - c) Curvilinear Cost
 - d) Fixed Cost
- 3) When graphing a mixed cost, what amount represents the Y-intercept?
 - a) Total Variable Cost
 - b) Variable Cost per unit
 - c) Mixed Cost
 - d) Fixed Cost
- 4) A budget is *most* likely to be effective if _____.
 - a) It is used to assess blame when things do not occur according to plans
 - b) It is not used to evaluate a manager's performance
 - c) Employees & managers at the lower levels do not get involved in the budgeting process
 - d) It has top management support
- 5) Which one of the following budgets would be prepared for a manufacturer but *not* for a merchandiser?
 - a) Direct Labor Budget
 - b) Cash budget
 - c) Sales budget
 - d) Budgeted Income Statement
- 6) Participative budgeting is a _____.
 - a) top-to-bottom approach
 - b) bottom-to-top approach
 - c) unrealistic in large companies
 - d) results in less-effective budgets
- 7) _____ are costs incurred to avoid poor-quality goods or services.
 - a) External failure costs
 - b) Appraisal costs
 - c) Internal failure costs
 - d) Prevention costs

- 8) Which of the following is *most* likely to be categorized as an external failure cost?
- a) The cost of machinery maintenance to avoid breakdowns during production
 - b) The cost of testing a product before shipment
 - c) The cost of lost future sales after a customer finds a defect in a product
 - d) The cost of inspection of materials to be used in production
- 9) When compared to the budget, if the actual cost or revenue contributes to higher income, the variance will be _____.
- a) Favorable
 - b) Unfavorable
 - c) Neither
 - d) Unable to determine from the information given
- 10) The flexible budget is prepared using the _____ output level.
- a) Budgeted
 - b) Actual
 - c) Wished for
 - d) Planned
- 11) The purchasing manager decides to purchase cheaper materials, which are of lower quality, in order to meet his department's budget expectations for the year. What unintended consequences will this likely have?
- a) The labor efficiency variance will likely be favorable.
 - b) The labor rate variance will likely be unfavorable.
 - c) The materials usage variance will likely be unfavorable.
 - d) The sales volume variance will likely be favorable.
- 12) The idea that managers should focus on all variances of significant amounts, both favorable and unfavorable, is called _____.
- a) Benchmarking
 - b) Ideal Standards
 - c) Standard Costing
 - d) Management by Exception
- 13) A static budget is based on a _____ amount of sales or production.
- a) Fixed
 - b) Variable
 - c) Changing
 - d) Potential

- 14) Practice of comparing a company's processes and performance standards against those of industry leaders is called _____.
a) Benchmarking
b) Variance Analysis
c) Standard Costing
d) Job Order Costing
- 15) Generally Accepted Accounting Principles allow a company to _____.
a) report Inventory at the standard cost but Cost of Goods Sold must be reported at actual cost
b) report both Cost of Goods Sold and Inventory at the actual cost
c) report both Cost of Goods Sold and Inventory at the standard cost, as long as, the difference is immaterial
d) report Cost of Goods Sold at standard cost, but Inventory must be reported at the actual cost

SHORT-ANSWER (4 points each)

Star Labs maintenance cost over the last six (6) months is below. Use the high-low method to estimate the fixed and variable components of the maintenance cost.

Month	Maintenance Hours	Maintenance Cost
July	18	\$6,919
August	12	5,134
September	15	6,007
October	21	6,951
November	24	8,267
December	6	3,470

- 1) Variable Cost per unit _____
- 2) Fixed Cost in total _____
- 3) If maintenance hours for the following January are expected to be 22, estimate the total maintenance cost for the month.

Vibe manufactures sunglasses that sell for \$180 per pair and whose variable costs are \$135 per pair. The annual fixed costs are \$575,000. Use this information to compute the following:

- 4) Contribution margin per unit _____
- 5) Contribution margin ratio _____
- 6) Break-even point in units _____
- 7) Break-even point in dollars of sales _____

Vibe adds a second version of sunglasses that sell for \$220 per pair and whose variable costs are \$155 per pair. Vibe expects the ratio to be 3:1 version 1 to version 2. Fixed costs are now \$625,000.

8) What is the weighted-average contribution margin per unit? _____

9) How many of the Version 1 sunglasses will be sold at break-even? _____

10) How many of the Version 2 sunglasses will be sold at break-even? _____

Based on plans to produce 24,000 units, Vibe budgeted fixed costs to be \$300,000 and variable costs to be \$264,000. If actual production was 20,000, what would the flexible budget report show for:

11) Fixed Costs? _____

12) Total Variable Costs? _____

PROBLEM 1: Budgeting Scenarios (100 points)

Near the end of 2018, Star Labs prepared the following estimated balance sheet for 2018.

Star Labs					
Estimated Balance Sheet					
December 31, 2018					
ASSETS			LIABILITIES & EQUITY		
Cash	\$431,000		Accounts Payable	\$65,000	
Accounts Receivable	105,000		Taxes Payable	10,000	
Inventory	<u>175,000</u>		Notes Payable (due in 2022)	<u>390,000</u>	
Total Current Assets		\$711,000	Total Liabilities		\$465,000
Equipment (net)		<u>472,500</u>	Common Stock	472,000	
Total Assets		<u>\$1,183,500</u>	Retained Earnings	246,000	
			Total Stockholders' Equity		<u>718,500</u>
			Total Liabilities & Equity		<u>\$1,183,500</u>

Star Labs currently sells only one product, which is purchased for \$35 per unit and resold for \$60. Management expects to sell 7,000 units in the first quarter, with sales increasing by 10% each month. Calculate the Sales Budget for each month in the first quarter.

Sales Budget (in dollars)	January	February	March

Management would like to have ending inventory equal to 20% of next month's sales. Calculate the Merchandise Purchases Budgets for each month in the first quarter.

Merchandise Purchases Budget (in dollars)	January	February	March

Sales Commissions equal to 20% of sales are paid each month. General and administrative expenses are \$144,000 per year and are paid evenly throughout the year. Maintenance expense is \$2,000 per month and Depreciation Expense is \$1,200 per month. Calculate the total amount of General, Selling, and Administrative Expenses for each month in the first quarter.

General, Selling, & Administrative Expenses	January	February	March

Management estimates that 30% of all sales are cash sales, and the remaining 70% of sales are on account. Of the credit sales, approximately 60% are collected in the month of sale, with the other 40% collected in the month after sale. Determine the expected cash collections for each month in the first quarter.

	January	February	March
Cash Collections			

Historically, management has paid for 20% of merchandise purchases in the month of purchase, and the remainder in the month following purchase. Determine the expected cash payments for the first quarter.

	January	February	March
Cash Payments			

Determine the total cash budget for the first quarter.

	January	February	March
Cash Budget			

Determine the balances in the following accounts at the end of the first quarter.

Accounts Receivable	
Accounts Payable	

PROBLEM 2: Variance Analysis (96 points)

West-Allen Manufacturing Corporation manufactures headphones. During the last quarter of 2018, 315,000 headphones were manufactured, and the following cost data was accumulated.

Standard Cost Information		
	Quantity	Cost
Direct Materials	2 parts	\$0.45 per part
Direct Labor	0.02 hours	\$8.00 per hour
Variable Manufacturing Overhead (Allocated based on Direct Labor Hours)	0.02 hours	\$10.00 per hour
Fixed Manufacturing Overhead (\$85,500 for static budget volume of 285,000 units and 5,700 hours, or \$15 per hour)		
Actual Cost Information		
Direct Materials	627,000 parts	\$332,310
Direct Labor	5,250 hours	\$43,050
Variable Manufacturing Overhead		\$56,700
Fixed Manufacturing Overhead		\$78,000

Compute the following variances and indicate whether each variance is Favorable (F) or Unfavorable (U).

Direct Materials Price Variance		Variable Overhead Cost Variance	
Direct Materials Usage Variance		Variable Overhead Efficiency Variance	
Total Materials Variance		Total Variable Overhead Variance	
Direct Labor Rate Variance		Fixed Overhead Cost Variance	
Direct Labor Efficiency Variance		Fixed Overhead Volume Variance	
Total Direct Labor Variance		Total Fixed Overhead Variance	